

Homework — 1.1.3 Input, Output and Storage Devices — ANSWER SHEET

Separate answer sheet / mark scheme — keep for marking.

Mark scheme

Part A (14 marks)

1. [3] — 1 mark each: Microphone = **input**; Laser printer = **output**; Sensor = **input**.
2. [2] — 1 mark: data is represented as a pattern of **pits and lands** burned into the disc surface (a spiral track). 1 mark: a **laser** is shone at the surface and the **reflected light** differs between pits and lands, which is interpreted as the binary data.
3. [3] — Advantages (1 mark each, max 2): faster read/write; no moving parts → more durable/shock-resistant; silent; lower power consumption. Disadvantage (1 mark): higher cost per GB; finite number of write cycles; (typically lower maximum capacity than HDD).
4. [3] — 1 mark recommendation: **magnetic tape** (accept HDD for cost reasons). Up to 2 marks justification linked to the scenario: very **low cost per GB / cheap for huge capacity**; suited to **archive/backup** that is **rarely accessed** (serial access is acceptable when random access is not needed); long shelf life / removable for off-site storage. (*Reject generic pros not linked to "cheap rarely-accessed archive"; cap at recommendation mark.*)
5. [2] — 1 mark: **cache is faster** than RAM (RAM is slower). 1 mark: cache is **on/near the CPU** (small), whereas RAM is **separate main memory** further from the CPU (larger). (*Accept: CPU checks cache before RAM.*)
6. [1] — RAM is full, so the OS is using **virtual memory** — moving pages out to the **disk (paging/swap file)** and back; the constant disk paging (disk thrashing) is what makes it slow.

Part B (6 marks)

7. [2] — 1 mark per difference (max 2): RISC few simple fixed-length instructions vs CISC many complex variable-length; complexity in software/compiler (RISC) vs hardware (CISC); RISC easier to pipeline; RISC lower power; ARM mobile/embedded vs x86 desktops.
8. [2] — 1 mark: a GPU has **thousands of small cores**. 1 mark: so it applies the **same operation to large blocks of data in parallel** (data parallelism), giving far higher throughput than sequential processing.
9. [2] — 1 mark: **Memory Data Register**. 1 mark: holds the **data or instruction** just fetched from memory, or about to be written to memory.

Total: 14 + 6 = 20